

YACHEN MIAO

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EDUCATION

University of Liverpool, Liverpool, UK

Sep 2025 - Jun 2027 (expected)

Bachelor of Science in Applied Mathematics (2+2 Programme)

- Average: 93/100 (Ranked 1st, Year 2); First-Class Honours Expected
- Relevant Coursework: Statistics & Probability I & II, Financial Mathematics, Numerical Methods;
In Progress: Theory of Statistical Inference, Linear Statistical Models, Measure Theory and Probability

Xi'an Jiaotong-Liverpool University, Suzhou, China

Sep 2023 - Jun 2025

Bachelor of Science in Applied Mathematics

- Average: 86/100 (Top 5%, Year 1); 72/100 (Foundation Year, Year 0)
- Relevant Coursework: Analysis 1 & 2, Introduction to the Methods of Applied Mathematics, Linear Algebra, Introduction to Programming in Java

AWARDS & SCHOLARSHIPS

- Willis Prize in Mathematics (Ranked 1st, Year 2 Applied Mathematics) 2026
- XJTLU Excellence Scholarship (Top 3, Year 1 Applied Mathematics) 2025
- Continuing Academic Merit Scholarship 2024
- First Class Entrance Scholarship 2023

RESEARCH EXPERIENCE

Distortion Risk Measures and Stochastic Dominance, Liverpool, UK

Jun 2026 - Present

Independent Study under Faculty Guidance

- Studied distortion risk measures through their distributional and quantile representations, working with distribution functions, generalised inverses and integral formulations.
- Examined the technical conditions required in proof-based arguments, including cases where quantile functions diverge as the probability level approaches one and additional tail or integrability conditions may be needed.
- Preparing to extend this work towards fractional stochastic dominance with a faculty member at Liverpool, with the possibility of developing it into a final-year research project.

SELECTED PROJECTS

Finite Memory and Generalisation in Sequence Models

Oct 2025 - May 2026

Independent Research Project

- Built character-level finite-order Markov models on a Shakespeare corpus and evaluated orders 1–8 using held-out negative log-likelihood.
- Found validation performance peaked at order 3 as higher-order contexts became increasingly sparse, illustrating the trade-off between model flexibility and finite-sample estimation.
- Investigated how fixed-memory assumptions restrict sequence modelling and used attention-based models as a conceptual comparison for unrestricted context dependence.

Quantitative Finance Project, University of Liverpool

Apr 2026 - May 2026

Option Pricing, Risk Estimation and CAPM Analysis

- Developed a Monte Carlo framework for option valuation, using 10,000 simulated paths to quantify pricing uncertainty and assess convergence of the estimated option value.
- Constructed a 95% confidence interval for the option price, explicitly characterising simulation uncertainty rather than relying on a single point estimate.
- Estimated systematic risk from historical return data, deriving Lloyds Bank's market beta ($\beta \approx 1.34$) through covariance based estimation and examining its implications under CAPM.

Unilever Mathematics Challenge, Team Leader

Jan 2026 - Feb 2026

- Led a 6-member team analysing Unilever shampoo prototype data; nonlinear patterns in the data led me to use tree-based ensembles instead of a linear model.
- Developed a standardised sustainability scoring framework combining cost, environmental and consumer-preference measures to compare competing formulations consistently.

PROFESSIONAL EXPERIENCE

Shanghai Tianyidao Investment Management Co., Ltd., Shanghai, China

Jun 2026 - Aug 2026

Quant Research Intern

- Applied Benjamini-Hochberg FDR control to manage multiple testing across large numbers of candidate factors.
- Designed a two stage validation gate for AI-generated factors, requiring strong in-sample performance to persist under out-of-sample robustness and stability tests.
- Made substantial code contributions the FactorLab codebase, integrating automated factor generation, replication and statistical evaluation into an end-to-end research workflow.

TECHNICAL SKILLS & INTERESTS

Technical Skills: Python; R; Stata; MATLAB; Maple; SQL; Tableau; Excel

Tools: Git/GitHub, Jupyter, VS Code, LaTeX, pytest, Linux/WSL2

Interests: Strength Training; Investing; Photography